

National Neighborhood Data Archive (NaNDA): Grocery Stores by Census Tract, United States, 2003-2017



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Overview and Data Dictionary

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Dataset Overview

Description

This dataset contains measures of the number and density of grocery stores – including supermarkets, specialty food stores, and warehouse clubs – per United States census tract from 2003 through 2017. These types of businesses represent places where neighborhood residents can obtain fresh and healthy foods.

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Data Sources

The North American Industry Classification System (NAICS) was used to identify organizations that can be considered grocery stores. NAICS was developed by the United States Office of Management and Budget and is “the standard used by Federal statistical agencies in classifying business establishments for the purpose of collecting, analyzing, and publishing statistical data related to the U.S. business economy” (United States Census Bureau, 2017).

Establishment data was taken from the National Establishment Time Series (NETS) database, which provides annual records of the US economy for more than 60 million private for-profit and nonprofit establishments and government agencies (Walls & Associates, 2017). This longitudinal data source classifies establishments by NAICS code and provides detailed business microdata and address history. NETS is produced in most years by Walls and Associates using archival establishment data from Dun & Bradstreet. While it currently includes data from 1990 to 2015 and 2017, only data from 2003-2015 and 2017 were used to create this dataset. (NETS data was not produced for 2016.)

Census tracts for establishments were determined using the 2010 TIGER/Line shapefiles produced by the United States Census Bureau (United States Census Bureau, 2010).

Census tract population figures for 2000-2010 were taken from Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019). Population figures for 2011-2017 were obtained from Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020).

Coverage

The dataset contains one observation per census tract per year in the United States, Puerto Rico, and the US Virgin Islands. Other US island territories are excluded.

Methodology

The purpose of this dataset is to measure the number and density of businesses where people might obtain fresh, healthy, high-quality food in their neighborhood. Prior research has demonstrated that people living in neighborhoods with limited access to fresh food are at higher

risk of adverse health conditions such as obesity, diabetes, and atherosclerosis (Morland & Everson, 2009, Auchincloss et al., 2008, Morland et al., 2006). In particular, “full-line” grocery stores, as opposed to other stores that sell food such as convenience stores, are an important source for fresh, healthy, affordable food in one’s neighborhood (Gomez-Lopez et al., 2017, Mantovani et al., 1997). Access to supermarkets is limited for residents of predominantly minority neighborhoods, impoverished neighborhoods, and rural areas (Zenk et al., 2005, Larson et al., 2009). Grocery stores are also significant as neighborhood destinations—along with community resources such as libraries, parks, gyms, and coffee shops—that foster social connection, especially for older adults and in impoverished neighborhoods (Finlay & Kobayashi, 2018, Hickman, 2013).

Our research team identified the following NAICS codes as representing the types of businesses selling fresh, healthy foods that people can prepare and eat at home:

- 445110: supermarkets and grocery stores, excluding convenience stores.
- 4452: all specialty food stores. These include meat, fish, and seafood markets, produce markets, baked goods stores, and spice stores.
- 452311: warehouse clubs and supercenters, which sell fresh food alongside canned and packaged foods and other types of merchandise (such as clothing and furniture).

For each NAICS code we identified, we queried the NETS database for all establishments that were open at some time between 2003 and 2015 and in 2017 (NETS does not contain data for 2016), and their location during each year. Note that NETS data were not cleaned or verified during this process, and that NETS data has limitations and inaccuracies as described in Gomez-Lopez et al. (2017) and Finlay et al. (2019).

We determined each establishment’s census tract for each year by mapping its latitude and longitude provided in the NETS database to the 2010 version of the TIGER/Line shapefiles from the US Census Bureau.

Using these locations, we created three variables to represent establishment counts per tract. The first count variable is the total of all establishments with that NAICS code in the tract. The second is the count of just establishments with non-zero sales figures in the current year, which can be used where desired to filter out establishments with very low business volume. The third is the count of establishments employing more than one person, which can be used to filter out very small owner-operated businesses if desired. The most useful variable for a given research question may vary by establishment type, so we have created three count types for all NAICS codes so that users may make their own decision about the version that best meets their needs.

We then calculated population densities (per 1000 people) for each of the three count variables for each NAICS code. Tract populations for 2003-2009 come from the NaNDA dataset [Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010](#). Values for each year are interpolated. For 2010-2012 and 2013-2017, we used the tract population from the NaNDA dataset [Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017](#). Population figures in this dataset

were taken from the 2012 and 2017 ACS five-year estimates, respectively, in order to avoid overlapping ACS estimates as advised by the Census Bureau, and for compatibility with other NaNDA datasets. Population figures for 2010-2017 are not interpolated. Consult the codebooks for the NaNDA datasets Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019) and Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020) for details on why these population figures were selected and how 2003-2009 population was interpolated.

We also created three variables representing the density of establishments per square mile. The area is the land area of the tract from the 2010 version of the TIGER/Line shapefiles. As a result, all area densities are standardized to 2010 census geography and are comparable across years across the entire dataset.

Population density was set to -1 for tracts with zero population in a given year. Population and area densities are missing for tracts in Puerto Rico and the US Virgin Islands, which are not included in the TIGER/Line shapefiles. Population and area densities are also missing in certain years for a few census tracts whose FIPS codes changed at some point between 2003 and 2017.

Finally, we merged counts and densities of all selected NAICS codes to create an aggregate dataset containing tract population, area, and count and density variables for all grocery stores per tract in each year.

Usage Notes

The businesses in this dataset were selected to represent locations where people can obtain fresh and healthy foods for consumption at home, and to inform the study of neighborhood disparities in spatial distribution of healthy food stores. This choice necessarily limits how well this dataset represents a neighborhood's overall food environment. Of course, we recognize that not all foods sold in grocery stores is fresh or healthy, that people also obtain food from other businesses (such as convenience stores and restaurants), and that people also consider factors besides health—such as taste, convenience, and cost—when deciding where to purchase food and what food to purchase.

Data users interested in a broader picture of a neighborhood's food landscape might find useful data in these other NaNDA datasets:

- [Dollar stores by census tract, United States, 2003-2017](#) (Gomez-Lopez et al., 2020).
- [Eating and drinking places by census tract, United States, 2003-2017](#) (Esposito et al., 2020).
- [Liquor, tobacco, and convenience stores by census tract, United States, 2003-2017](#) (Finlay et al., 2020a).

Data users interested in third places and community resources might find further useful data in these additional NaNDA datasets:

- [Arts, entertainment, and recreation establishments by census tract, United States, 2003-2017](#) (Finlay et al., 2020a), which includes gathering places such as gyms, libraries, museums, arcades, bowling alleys, and golf courses, among others.
- [Religious, civic, and social organizations by census tract, United States, 2003-2017](#) (Finlay et al., 2020d), which includes community groups such as churches, mosques, synagogues, ethnic associations, and veterans' associations, among others.
- [Retail establishments by census tract, United States, 2003-2017](#) (Finlay et al., 2020e), which includes stores selling furniture, clothing, sporting goods, home and garden supplies, books, and used merchandise, among other items.
- [Post offices and banks by census tract, United States, 2003-2017](#) (Finlay et al., 2020c).
- [Personal care services and laundromats by census tract, United States, 2003-2017](#) (Esposito et al., 2020b).
- [Social services by census tract, United States, 2003-2017](#) (Finlay et al., 2020f), which includes youth and senior centers.

Variables

For additional information on the definition of each NAICS code, see the Methodology section of this document and the 2017 NAICS website: <https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017>.

Variable	Type	Obs	Unique	Mean	Min	Max	Label
tract_fips10	string	1036462	74033	.	.	.	Census tract FIPS code (2010 TIGER/Line shapefiles)
year	int	1036462	14	2009.571	2003	2017	Year
population	float	1020881	250271	4208.894	0	65528	Tract population
aland10	float	1022434	72534	48.07265	0	85425.73	Tract land area, square miles
count_445110	int	1036462	55	2.300376	0	62	# supermarkets/grocery stores
count_sales_445110	int	1036462	50	2.133618	0	61	# supermarkets/grocery stores w/ sales>0
count_emp_445110	int	1036462	55	2.157833	0	61	# supermarkets/grocery stores w/ 2+ employees
popden_445110	float	1015637	287193	2.420655	-1	289507.3	# supermarkets/grocery stores per 1000 people
popden_sales_445110	float	1015610	278534	2.217033	-1	289507.3	# supermarkets/grocery stores per 1000 people w/ sales>0
popden_emp_445110	float	1015620	278791	2.376839	-1	289507.3	# supermarkets/grocery stores per 1000 people w/ 2+ employees
aden_445110	float	1017996	192314	4.840968	0	806.2399	# supermarkets/grocery stores per sq mile
aden_sales_445110	float	1017996	187195	4.470665	0	767.8475	# supermarkets/grocery stores per sq mile w/ sales>0
aden_emp_445110	float	1017996	187972	4.537774	0	767.8475	# supermarkets/grocery stores per sq mile w/ 2+ employees
count_4452	int	1036462	83	1.76452	0	137	# specialty food stores
count_sales_4452	int	1036462	77	1.599901	0	122	# specialty food stores w/ sales>0
count_emp_4452	int	1036462	79	1.540207	0	130	# specialty food stores w/ 2+ employees
popden_4452	float	1015648	242474	1.802188	-1	289507.3	# specialty food stores per 1000 people
popden_sales_4452	float	1015616	231913	1.648613	-1	289507.3	# specialty food stores per 1000 people w/ sales>0
popden_emp_4452	float	1015596	225294	1.717309	-1	289507.3	# specialty food stores per 1000 people w/ 2+ employees
aden_4452	float	1017996	169074	3.008285	0	804.8346	# specialty food stores per sq mile
aden_sales_4452	float	1017996	163814	2.760393	0	774.4635	# specialty food stores per sq mile w/ sales>0
aden_emp_4452	float	1017996	160276	2.671628	0	730.4467	# specialty food stores per sq mile w/ 2+ employees
count_452311	int	1036462	7	0.0278331	0	6	# warehouse clubs/supercenters

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Variable	Type	Obs	Unique	Mean	Min	Max	Label
count_sales_452311	int	1036462	7	0.0262827	0	6	# warehouse clubs/supercenters w/ sales>0
count_emp_452311	int	1036462	7	0.0264969	0	6	# warehouse clubs/supercenters w/ 2+ employees
popden_452311	float	1015101	13856	0.7592698	-1	289507.3	# warehouse clubs/supercenters per 1000 people
popden_sales_452311	float	1015096	13091	0.757916	-1	289507.3	# warehouse clubs/supercenters per 1000 people w/ sales>0
popden_emp_452311	float	1015090	13218	0.75185	-1	289507.3	# warehouse clubs/supercenters per 1000 people w/ 2+ employees
aden_452311	float	1017996	3059	0.020063	0	32.98298	# warehouse clubs/supercenters per sq mile
aden_sales_452311	float	1017996	3045	0.0188508	0	32.98298	# warehouse clubs/supercenters per sq mile w/ sales>0
aden_emp_452311	float	1017996	2973	0.0188499	0	32.98298	# warehouse clubs/supercenters per sq mile w/ 2+ employees

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